

PROJECT REPORT

MovieIQ – Predictive Analytics on Film Success

Abstract

MovieIQ is a machine learning project developed to predict whether a movie will be successful based on its characteristics. In this project, a movie is considered successful if its revenue is greater than its budget. The project includes data preparation, exploratory data analysis, statistical testing, predictive modeling using the Random Forest algorithm, and the development of an interactive Streamlit dashboard. The dashboard allows users to explore movie data, apply filters, and predict the success of a movie using a trained machine learning model.

1. Introduction

The film industry invests significant amounts of money in producing movies. Predicting whether a movie will be successful before its release helps studios and investors make informed decisions.

MovieIQ analyzes movie data using machine learning techniques to estimate whether a movie is likely to become successful.

2. Problem Statement

The objective of this project is to predict whether a movie will be successful based on its features.

A movie is labeled as:

- **Success = 1** if Revenue > Budget
- **Success = 0** if Revenue ≤ Budget

This is a binary classification problem because the output contains only two classes: Successful and Unsuccessful.

3. Objectives

The objectives of this project are:

- Analyze movie data
- Perform data cleaning and preprocessing
- Conduct exploratory data analysis

- Perform statistical testing
- Train a Random Forest classification model
- Predict movie success
- Build an interactive Streamlit dashboard

4. Dataset Description

The dataset contains approximately **2000 movies**.

Dataset Features

Feature	Description
Budget	Production budget
Revenue	Total movie revenue
Popularity	Popularity score
Runtime	Movie duration
Vote Average	Average audience rating
Title	Movie title
Genres	Movie genres
Target Variable	

Success

- 1 → Successful
- 0 → Unsuccessful

5. Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn

- SciPy
- Scikit-learn
- Streamlit
- Joblib
- Jupyter Notebook

6. Data Preparation

The following preprocessing steps were performed:

- Loaded the dataset
- Checked dataset dimensions
- Generated summary statistics
- Checked missing values
- Verified zero values in Budget and Revenue
- Removed invalid records if necessary
- Created the Success target variable
- Processed multiple genres into a usable format
- Created an exploded dataset for genre analysis

7. Exploratory Data Analysis

Several visualizations were created to understand the data.

7.1 Budget vs Revenue

A scatter plot was used to analyze the relationship between budget and revenue.

Observation

- Correlation = **0.76**
- Movies with larger budgets generally earned higher revenue.

7.2 Genre Analysis

Genre frequency and success rates were analyzed.

Most Common Genres

Genre	Movies
Romance	220
Adventure	207
Science Fiction	205
Animation	203
Comedy	202

Genre Success Rates

Genre	Success Rate
Drama	82.23%
Horror	81.91%
Animation	81.28%
Comedy	81.19%
Thriller	80.93%
Romance	80.91%
Adventure	80.19%
Action	78.65%
Science Fiction	78.54%

7.3 Popularity, Runtime and Vote Average

Average values were compared for successful and unsuccessful movies.

Feature	Success	Failure
Popularity	50.73	47.40
Runtime	129.08	131.62
Vote Average	6.02	6.12

Observation

Popularity showed the strongest relationship with movie success.

7.4 Correlation Heatmap

A correlation heatmap was created for all numerical variables.

Observation:

- Budget and Revenue showed a strong positive correlation.
- No serious multicollinearity was observed among the selected model features.

8. Statistical Testing

8.1 T-Test

Feature Tested:

Popularity

Hypotheses

H_0 : Mean popularity is the same for successful and unsuccessful movies.

H_1 : Mean popularity is different.

Results

T Statistic = **2.0617**

P Value = **0.039682**

Conclusion

Since $p < 0.05$,

The null hypothesis was rejected.

Popularity differs significantly between successful and unsuccessful movies.

8.2 Chi-Square Test

Feature Tested:

Genre

Results

Chi-Square Statistic = **1.7356**

P Value = **0.988072**

Degrees of Freedom = **8**

Conclusion

Since $p > 0.05$,

There is no statistically significant association between movie genre and movie success.

9. Machine Learning Model

Algorithm Used:

Random Forest Classifier

Selected Features

- Budget
- Popularity
- Runtime
- Vote Average

Excluded Features

- Revenue
- Title
- Genres

Revenue was excluded because it directly determines the target variable and would cause target leakage.

10. Train-Test Split

The dataset was divided into:

- Training Data → 80%
- Testing Data → 20%

Stratified sampling was used to preserve class proportions.

11. Model Performance

Metric	Value
--------	-------

Accuracy	80.5%
----------	-------

Precision	80.7%
-----------	-------

Recall	99.69%
--------	--------

Confusion Matrix

Actual / Predicted	Failure	Success
--------------------	---------	---------

Failure	0	77
---------	---	----

Success	1	322
---------	---	-----

Observation

The model predicts successful movies very well but struggles to correctly identify unsuccessful movies because the dataset is imbalanced.

12. Feature Importance

Feature	Importance
---------	------------

Popularity	26.21%
------------	--------

Budget	25.89%
--------	--------

Vote Average	25.57%
--------------	--------

Runtime	22.32%
---------	--------

Popularity was identified as the most influential feature.

13. Streamlit Dashboard

The MovieIQ dashboard includes:

- Dataset overview
- KPI cards
- Genre filter
- Minimum vote average filter
- Budget vs Revenue visualization
- Popularity analysis
- Statistical test results
- Movie success prediction
- Interactive interface

14. Deployment

The project was deployed using:

- GitHub Repository Link: <https://github.com/yashwanthreddy1845-oss/MovieIQ-Predictive-Analytics>
- Streamlit Community Cloud Link : <https://movieiq-predictive-analytics-dnisaet6fktbdclw9kkd9x.streamlit.app/>

The deployed application allows users to interact with the model directly through a web interface.

15. Limitations

- Dataset is imbalanced.
- Only four numerical features were used for prediction.
- Genre information was not encoded into the model.
- External factors such as cast, director, marketing budget, and release season were unavailable.

16. Future Improvements

Future improvements include:

- Use larger movie datasets.
- Handle class imbalance using SMOTE or similar techniques.
- Include categorical feature encoding.
- Compare Random Forest with XGBoost and Gradient Boosting.
- Add additional features such as actors, directors, production companies, and release dates.

17. Conclusion

MovielQ successfully demonstrates how machine learning can be used to predict movie success. The project integrates data preprocessing, visualization, statistical analysis, predictive modeling, and deployment into a complete end-to-end application.

The Random Forest model achieved an accuracy of **80.5%**, with popularity identified as the most influential feature. While the model performs well at identifying successful movies, improving performance on unsuccessful movies remains an area for future work.